

PRACTICAL NO : 04

Aim : Create a Terrain and add the following components :

- a) Mountain
- b) Snow
- c) Green Land
- d) sand
- e) Boat

Step 1

Download the Unity Hub

Before you can start creating in Unity you'll need to download and install the Unity Hub.

<http://unity3d.com>

Step 2

Install the Unity Hub

Once your download and install has completed, open the Hub and login or create a Unity account.

STEP 3

Install the Unity Engine

In the Hub, start a tutorial or open a new project. The latest version of the Unity Engine will download automatically.

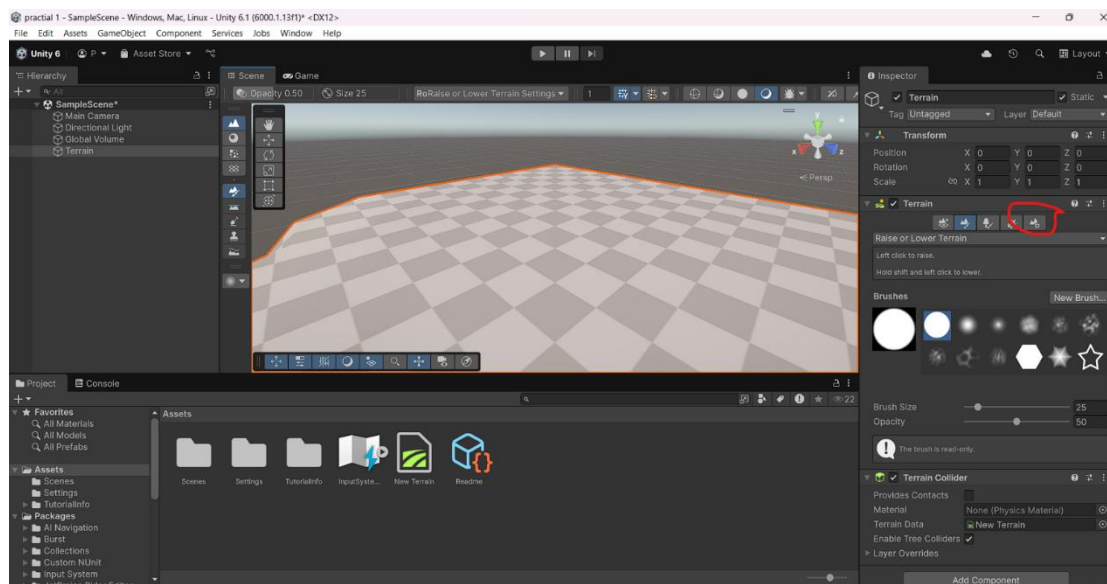
- Click on this link
- Scroll and click on learn more under game section
- Click on download Unity then click on download for windows option.
- Create new project with the proper project name and wait (as it will take some moment to open)

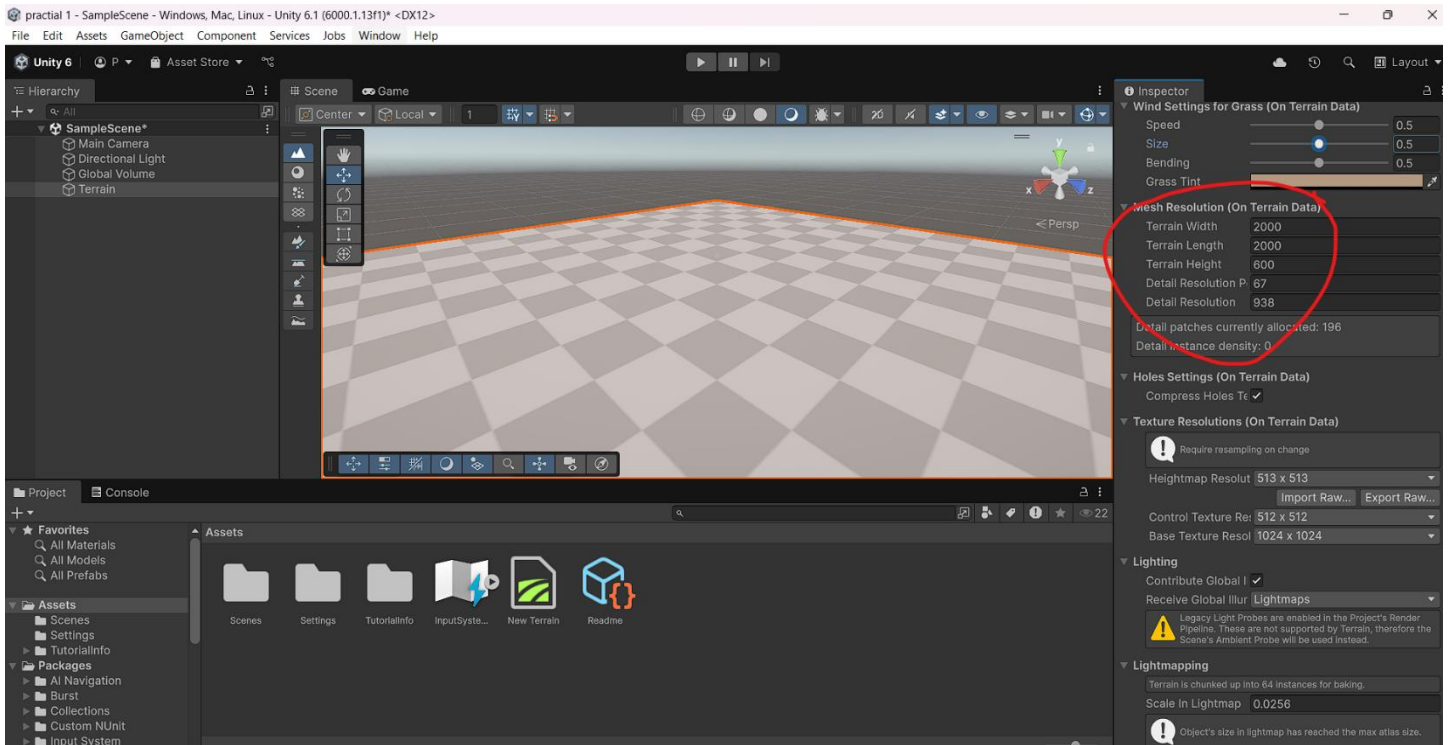
1) Create project & scene

1. Open Unity Hub → **New** → choose **3D (or URP 3D)** template → name it and create.
2. Save the scene: **File** → **Save As...** (e.g., IslandScene).

2) Create the Terrain

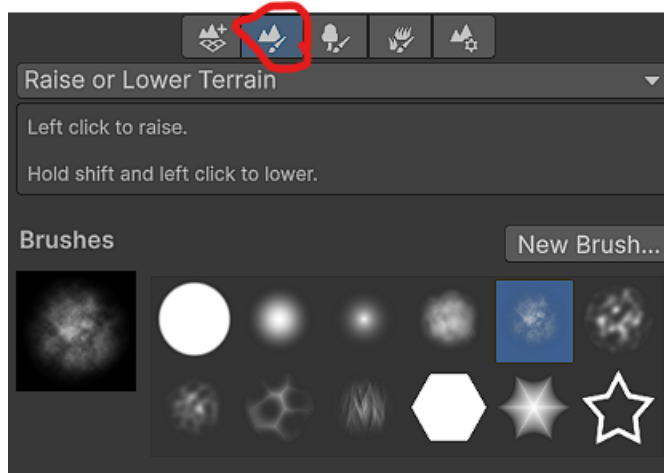
1. **GameObject** → **3D Object** → **Terrain**. Unity adds a Terrain GameObject and a TerrainCollider.
2. With the Terrain selected, open **Terrain Settings** (gear icon in Terrain Inspector).
 - Set **Width / Height / Length** (example: 2000 x 600 x 2000 where Height = max elevation).





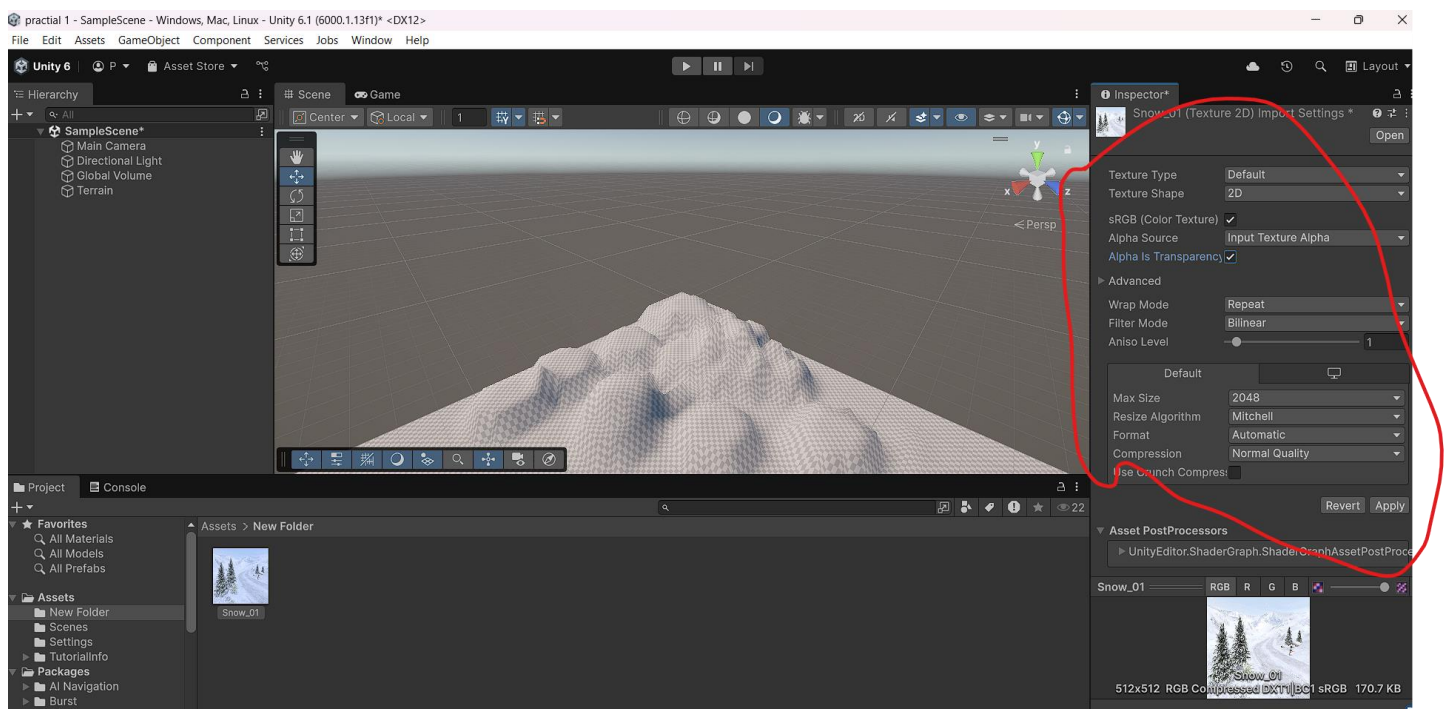
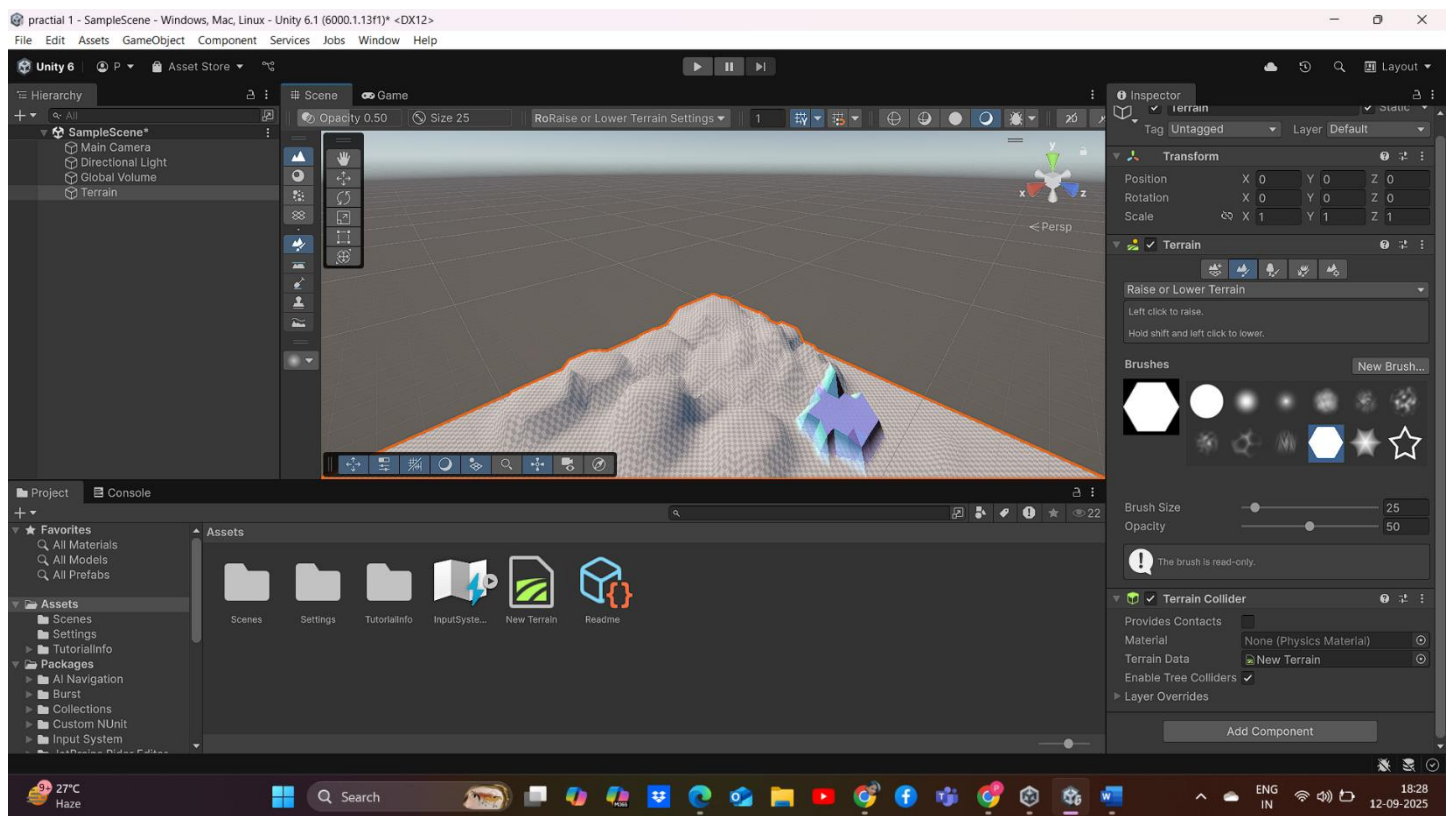
3) Sculpt mountains (basic)

1. Select Terrain → **Paint Terrain** (brush icon) → choose **Raise/Lower Terrain**.
2. Use a large brush to raise broad mountain shapes. Hold **Shift** while dragging to *lower*.



4) Prepare textures (snow, grass, sand)

1. Get or import texture images for: **grass/green**, **snow**, **sand**, and optionally **rock** (Assets → Create → Folder → drag textures in).
2. Select each texture in Project window → in Inspector:
 - **Texture Type:** Default.
 - **Wrap Mode:** Repeat.
 - Apply.



5) Add Terrain Layers (paint textures)

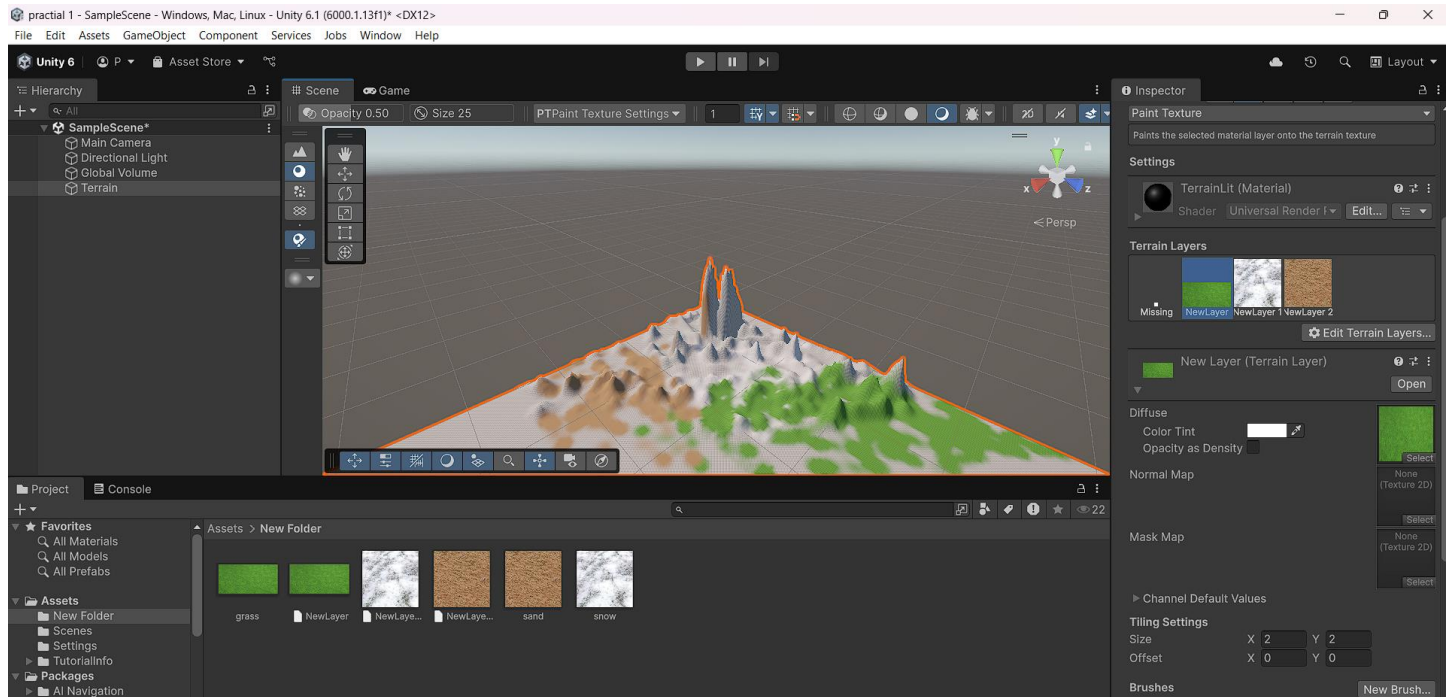
1. Click on your Terrain.
2. In Inspector → click the Paint Terrain (brush icon).
3. Select Paint Texture.
4. In the Terrain Layers section → click Edit Terrain Layers → Create Layer.
 - Pick your Grass texture.
5. Repeat the same to add Sand, Snow, etc.

6) Paint

1. Select the layer (e.g., Grass).
2. Pick a **brush shape** and adjust size/opacity.
3. Left-click and paint on the terrain!
 - Grass → lower areas.
 - Sand → near edges (where water will be later).
 - Snow → on higher peaks.



(USE THIS IMAGE OR DOWNLOAD FROM GOOGLE OF YOUR OWN)

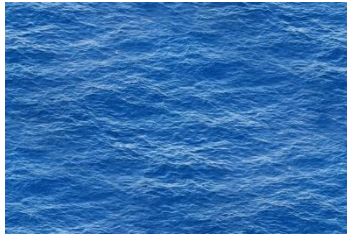


7 Add Water (for the boat + sand beach to make sense)

1. Go to **GameObject** → **3D Object** → **Plane**.
2. Rename it **Water**.
3. Scale it big (e.g., X = 200, Z = 200). Place it slightly lower than your sand area (like sea level).
4. Make a **new Material** (Right-click in Project → Create → Material → name WaterMat).
5. Select your water material (watermat).
In Inspector → make these changes.
 - Workflow Mode → set to Specular (not Metallic).
 - Surface Type → Transparent.
 - Blending Mode → Alpha.
 - Preserve Specular → ☒ keep checked.
 - Render Face → Both.
 - Scroll to Surface Inputs:
 - Base Map → choose a lighter blue color (not dark navy).
 - Smoothness → set higher (~0.8–1) → makes it shiny like water.
6. Drag this WaterMat onto your Water plane.

8) Steps to add water image in Unity

1. **Download a water texture** (you can find free ones from sites like cc0textures, polyhaven, or just search “free seamless water texture”).
 - Make sure you download **2 files** if possible:
 - **Albedo (color image of water)**
 - **Normal Map (bumpy/wavy effect)**
2. **Import into Unity**
 - Drag the images into your Unity project (Assets folder).
3. **Apply to Material (watermat)**
 - Select your **watermat material**.
 - In the **Base Map**, drag the **1st water image**.
 - In the **Normal Map** slot, drag the **second map image**.
 - If Unity asks “Fix Now?” → click it (to mark it as a normal map).



10) Create the Cube (Boat)

- In Hierarchy → Right-click → 3D Object → Cube.
 - Rename it → Boat.
- ☐ **Make it Boat-shaped**
- In the Inspector → Transform → Scale:
 - X = **15** (length)
 - Y = **0.5** (height)
 - Z = **15** (width)
- ☐ **Move it on the Water**
- In Inspector → Transform → Position:
 - Y should be just a little above the water plane (like 0.3 if your water is at 0).
 - If it disappears → it might be under the water → just drag it up in the Scene view.
- ☐ **Add Color / Material**
- Right-click in Assets → Create → Material → name BoatMat.
 - In Inspector → Base Map color → choose yellow or red.
 - Drag this material onto the cube.

